

Curriculum Intelligence Engine — Week 5: Evaluation & Error Analysis

Group 9 · Roll Nos: J048, J049, J050, J052, J053 · AI University NLP Course Project 2026 · 9 Oct 2026

CODE

```
Repository : https://github.com/jainamsethia/Curriculum-Intelligence-Engine
Tag        : week-5-redo
Commit     : 85f3791d2804d1fd1a55c371cd156e2121d902b7
Run        : python -m curriculum_engine.run all (data → week2 → week3 → week4 → week5; NMIMS_ZIP=<path to "B TECH.zip">)
This week  : curriculum_engine/experiments/week5.py, reports/failure_notes.json
Folders    :
.gitignore
README.md
curriculum_engine/ the NLP engine (parser, data, overlap, redundancy, prerequisites, diff, Bloom, evaluation)
labels/           LLM-annotated reference labels (ids + labels per pass + consensus)
legacy/           old Week 3-4 scripts and reports (history only)
legacy_mit/       old MIT OCW scripts (not used)
reports/          report templates (numbers are placeholders)
requirements.txt
results/          metrics JSON + figures per week
submissions/      per-week submission folders
tests/            unit tests (pytest)
tools/            report builder, report checker, submission packer
```

Feedback addressed

Instructor comment	What we changed	Evidence
Week 4: "overlapping content, missing pre-requisites, detect redundancies"	The evaluation is organised around these three capabilities first, each against a target fixed in the proposal.	results/week5/final.json
Week 4: "Where is the code?"	CODE block, tag, code zip; one command reruns every number in this report.	curriculum_engine/run.py
Week 2: "value to NMIMS"	Breakdown by NMIMS programme, failure cases on real NMIMS courses.	results/week5/per_programme.json

Progress vs previous week

Item	Week 4	Week 5 (this report)
Overlap method	ablations, selected on validation	final evaluation of fusion of TF-IDF and MaxSim (validation percentiles) on the held-out test split
Evidence of robustness	none	perturbations of the input text
Error analysis	none	12 failure cases with root cause and fix
Targets checked	none	13 of 14 met, 1 missed

1. Test results against the proposal targets

Held-out test split: course families never seen in training or tuning; thresholds come from validation. Overlap and flag precision use LLM-annotated reference labels (pass 1 Claude, pass 2 Phi-3 or a second Claude prompt, pass 3 on disagreements). No human annotators were used; results are indicative. For 94 of 200 pairs the second labeller was a Claude prompt rather than Phi-3, so agreement on those items is not independent. Detector flags (Week 3 onwards) were labelled by two Claude passes with different wording plus a pass 3, with no Phi-3, so agreement on flags is not independent.

Capability	Metric (test split)	Result	Target	Met?
1 Overlap	Spearman	0.717	≥ 0.6	yes
	ROC-AUC (any overlap)	0.965	≥ 0.85	yes
	F1 at validation threshold	0.731	≥ 0.7	yes
2 Redundancy	precision of top flags (reference labels)	0.800	≥ 0.7	yes
	injected near-duplicates in	1.000	≥ 0.8	yes

	top 20			
3 Prerequisites	removal recall	0.880	≥ 0.8	yes
	precision of flags (reference labels)	0.100	≥ 0.6	no
	false alarms after renaming	0.120	≤ 0.2	yes
4 Curriculum diff	precision (real year pairs)	0.998	≥ 0.85	yes
	recall (real year pairs)	0.956	≥ 0.85	yes
5 Bloom	accuracy vs faculty tags	0.776	≥ 0.7	yes
	within one level	0.871	≥ 0.85	yes
6 Emerging topics	precision of flagged gaps	1.000	≥ 0.7	yes
7 Topic labels	share accepted	0.700	≥ 0.7	yes

The one missed target is the precision of missing-prerequisite flags: most flags concern prerequisites that are taught earlier under another name, umbrella names such as "Engineering Mathematics", or phrases that are not subjects at all (cases F05–F07 below). Removal recall and rename false alarms meet their targets, so the checker finds real gaps but raises too many warnings to be used without review.

The overlap method has Spearman 0.717 [0.624, 0.794] and ROC-AUC 0.965 [0.937, 0.987] on 119 test pairs; at the validation threshold, precision 0.950 and recall 0.594.

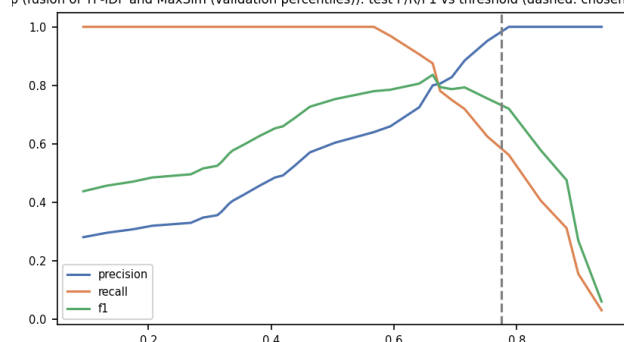
2. Breakdown by programme

Programme	Labelled pairs	Overlap ROC-AUC	Tagged outcomes	Bloom accuracy	Prerequisite cases	Removal recall
AI	4	n/a	23	0.696	1	1.000
CIVIL	15	1.000	19	0.842	16	1.000
COMMON	0	n/a	3	1.000	0	n/a
COMP	14	0.950	26	0.692	9	1.000
CSBS	26	0.990	13	0.692	12	0.833
CSE-CYBER	8	1.000	18	0.722	2	1.000
CSE-DS	18	0.944	18	0.778	5	0.800
DS	19	0.960	40	0.775	3	1.000
ELECTRICAL	4	n/a	0	n/a	0	n/a
EXTC	20	0.933	35	0.743	10	0.800
IT	14	0.875	23	0.696	2	1.000
MECHANICAL	36	0.975	14	0.857	14	0.714
MECHATRONICS	15	1.000	14	0.857	9	0.889

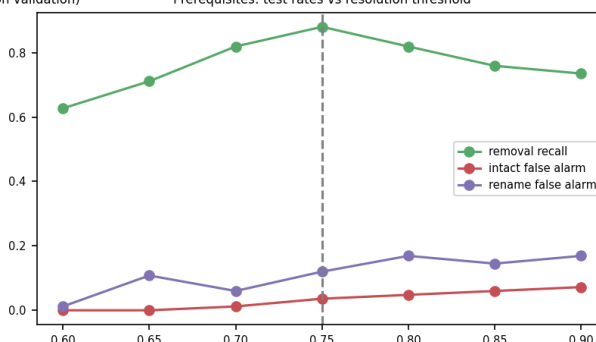
Programmes with few labelled pairs get no AUC (shown as n/a). Bloom accuracy uses the verb lexicon against faculty tags; removal recall is the share of removed prerequisite courses that raise a flag.

3. Threshold sensitivity

p (fusion of TF-IDF and MaxSim (validation percentiles)): test P/R/F1 vs threshold (dashed: chosen on validation)



Prerequisites: test rates vs resolution threshold



Left: overlap precision, recall and F1 on test as the threshold moves (dashed: the threshold chosen on validation). Right: prerequisite removal recall and false alarms as the resolution threshold moves.

At the validation threshold the test F1 is 0.731; the best F1 any threshold could reach on test is 0.836, a clear gap: a threshold tuned on 80 validation pairs is noisy, which is why the ranking metrics (Spearman, ROC-AUC) are the more reliable summary.

4. Robustness

Course B of every labelled test pair is perturbed and the pair is re-scored (left of the arrow: clean text).

Perturbation	Pairs	Spearman fusion of TF-IDF and MaxSim (validation percentiles)	Spearman TF-IDF	ROC-AUC fusion of TF-IDF and MaxSim (validation percentiles)	ROC-AUC TF-IDF
typos	119	0.717 → 0.717	0.708 → 0.709	0.965 → 0.965	0.959 → 0.959
word drop	119	0.717 → 0.706	0.708 → 0.702	0.965 → 0.958	0.959 → 0.955
shuffled units	119	0.717 → 0.717	0.708 → 0.709	0.965 → 0.965	0.959 → 0.959
truncated half	119	0.717 → 0.692	0.708 → 0.686	0.965 → 0.948	0.959 → 0.944
missing sections	119	0.717 → 0.707	0.708 → 0.666	0.965 → 0.958	0.959 → 0.931

Typos and shuffled units change nothing and dropping words very little; keeping only half of the units costs the most. Without outcomes and objectives TF-IDF loses more than the fusion, because MaxSim still matches the units. A perturbed copy of a course stays very close to the original (median self-similarity after typos 1.000, after truncation 0.988), so versions of one course are always separated from different courses: versions are a separate version-change task.

5. Failure cases

Selected by fixed rules, not hand-picked: that pass the threshold and the three lowest-scoring overlapping pairs below it; the first three prerequisite flags, two redundancy

flags and one emerging-topic gap the reference labels call wrong; the first topic the detector calls covered although the labels see a gap; and the two Bloom errors furthest from the faculty tag. Root causes and fixes: reports/failure_notes.json.

#	Type	Input	Prediction / truth	Category	Root cause	Fix or known limitation
F01	overlap false positive	Computer Networks vs Cloud Computing	score 0.781 (threshold 0.775) / no overlap	shared vocabulary, different subject	Both syllabi use networking-layer vocabulary (layers, protocols, services, virtualisation), so lexical and semantic scores pass the threshold although the subjects differ.	Weight matches by distinctive topic terms and require several matched units, not one high course score.
F02	overlap false negative	Unmanned Aerial Vehicles vs Engineering Mechanics	score 0.569 (threshold 0.775) / partial overlap	one shared unit diluted	Only one unit-level topic (rigid-body dynamics) is shared inside otherwise different courses, so the whole-course score stays below the threshold.	Report partial overlap from the unit-level detector (best unit pair) instead of the course score.
F03	overlap false negative	Cybersecurity for Non-IT vs Foundation of Information systems	score 0.594 (threshold 0.775) / partial overlap	one shared unit diluted	One security unit is shared; the rest of each course (IS strategy vs cryptography, IAM) pulls the course score down.	Same as above: unit-level evidence for partial overlap.
F04	overlap false negative	Cognitive Science & Analytics vs Advanced Business Analytics in Python and R	score 0.600 (threshold 0.775) / partial overlap	one shared unit diluted	Shared statistics topics (hypothesis testing, ANOVA) are phrased differently and sit inside larger,	Unit-level evidence; a domain-adapted encoder for statistics wording.

					different courses.	
F05	prerequisite flag judged wrong	Open Elective - I (BSC) - Fundamentals of Fuels and Lubricants (CIVIL) needs 'Chemistry (Semester- II)'	flagged as missing / reference labels: warning is wrong	noisy prerequisite text	The phrase keeps '(Semester- II)' and the first-year chemistry course has a different name, so neither the name nor the topic match passes.	Strip semester and code annotations from phrases; link common first-year courses to every programme explicitly.
F06	prerequisite flag judged wrong	Open Elective - I (BSC) - Modelling and Simulation (CIVIL) needs 'Engineering Mathematics'	flagged as missing / reference labels: warning is wrong	umbrella prerequisite name	'Engineering Mathematics' names a group of courses (Calculus, Linear Algebra, Mathematics - III); it resolves to a same-named course of another programme that Civil does not offer.	A small ontology of umbrella subjects mapped to their member courses.
F07	prerequisite flag judged wrong	Remote Sensing, GIS and Data Science Applications in Environmental Management (CIVIL) needs 'geography'	flagged as missing / reference labels: warning is wrong	not a course subject	A background phrase ('geography') is treated as a required course; no earlier unit is close enough to it.	Classify phrases as subject or background before checking (e.g. the LLM extractor).
F08	redundancy flag judged wrong	Design of Reinforced Concrete Structures vs Design of Steel Structures	flagged as redundant / reference labels: warning is wrong	same method, different object	Both courses teach limit-state design, so the unit texts share method vocabulary although the material differs (concrete vs steel).	Require overlap of the domain nouns, or a stricter duplicate threshold for design units.
F09	redundancy flag judged wrong	Introduction to Cryptography vs Elliptic Curve Cryptography	flagged as redundant / reference labels: warning is wrong	same topic, different depth	Public-key cryptography appears in both, as a survey in one and in depth in the other; the unit score cannot see depth.	Compare hours and outcome levels of matched units; report as partial, not redundant.
F10	missed emerging-topic gap	'Large language models and generative AI' in programme CSBS	judged as already taught / reference labels: a real gap	lenient threshold	The topic threshold reused from the prerequisite check lets weakly related units (code generation, RNNs) count as coverage.	Tune a separate threshold for emerging topics on labelled checks; this also explains the low gap recall.
F11	Bloom level error	Apply, Deploy, and Manage the applications efficiently while evaluating and integrating best practices in real-world sce	Apply / faculty tag Create	multi-verb outcome	The lexicon takes the first known verb ('apply'), but the outcome also asks to evaluate and integrate, which the faculty tag as Create.	Use the highest-level verb, or the classifier, when an outcome has several verbs.
F12	Bloom level error	Construct the curves using surveying instruments	Create / faculty tag Apply	domain-specific verb sense	'Construct' is Create in Bloom verb lists, but constructing survey curves is a practical skill the faculty tag as Apply.	A domain-specific verb map learned from faculty tags (needs more tagged outcomes).

6. Limitations

- Data: NMIMS semester syllabus booklets; 531 of 2,413 non-exam PDFs have a text layer and are parsed, the rest are scanned (OCR is future work), so programme structures and year-over-year comparisons have gaps.
- Labels: LLM-annotated reference labels; LLM-vs-LLM agreement, reported separately: Claude vs Phi-3 on 106 pairs 74% (weighted kappa 0.561); Claude vs Claude (wording B) on 94 pairs 99% (weighted kappa 0.969); the same model family wrote the code and pass 1, which is why label-free tests (injection, removal, renaming, faculty Bloom tags) run alongside.
- Synthetic tests use simple edits (word drop, typos, renamed titles); real revisions can be subtler. Robustness: paraphrase variant not run (time).

AI assistance

Code and report text were written with AI assistance (Claude). Reference labels are LLM-annotated (Claude, Phi-3). Every number is generated from `results/*.json` by `tools/build_report.py` and verified by `tools/check_reports.py`.